# Quantifying Co-evolution in Urban Systems with Transfer Entropy

CCS 2026

---

Juste Raimbault[1, 2, 3, 4, *]

13/10/2026



Géodata Paris

l'école des ingénieur·es géographes

[1]LaSTIG, Université Gustave Eiffel, IGN-Géodata Paris
[2]CASA, UCL
[3]UPS CNRS 3611 ISC-PIF
[4]UMR CNRS 8504 Géographie-cités
* juste.raimbault@geodata-paris.fr

# Co-evolution between networks and land-use





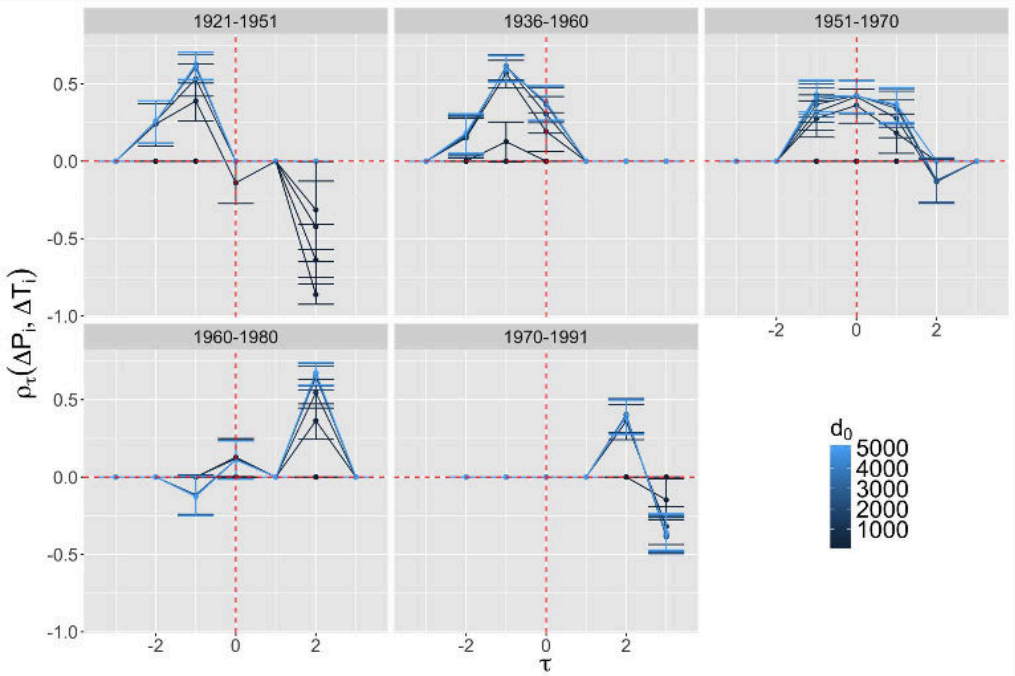



- Examples of various land-use impacts of new High Speed Rail stations [Blanquart and Koning, 2017]
- Transportation networks respond to land-use dynamics at other timescales, with governance processes involved [Raimbault and Le Néchet, 2021][Le Néchet and Raimbault, 2026]

# Characterising co-evolution in territorial systems

→ [Raimbault, 2021a] : a definition of co-evolution in territorial systems, transfered from ecology, based on statistical circular causalities on populations of objects

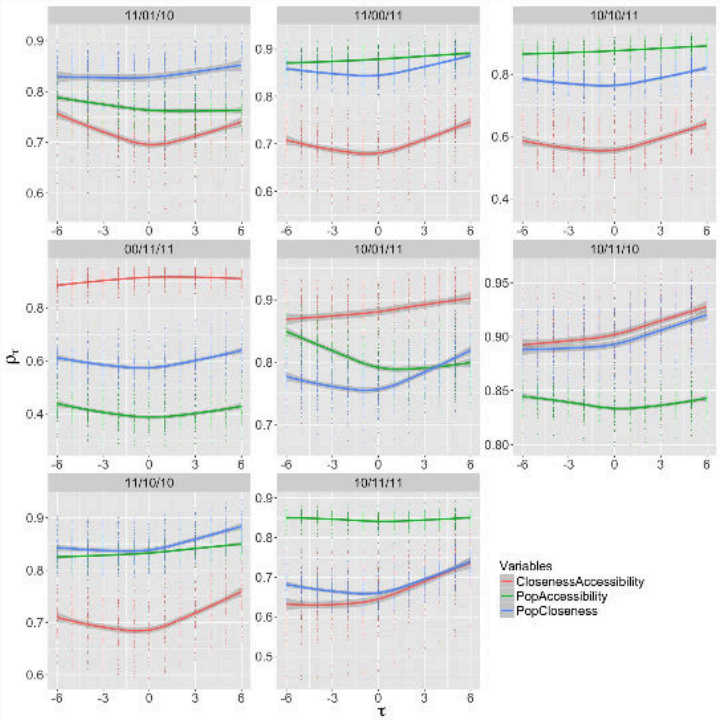
→ [Raimbault, 2017] : a method based on lagged correlations (Granger causality) to identify "causality regimes"



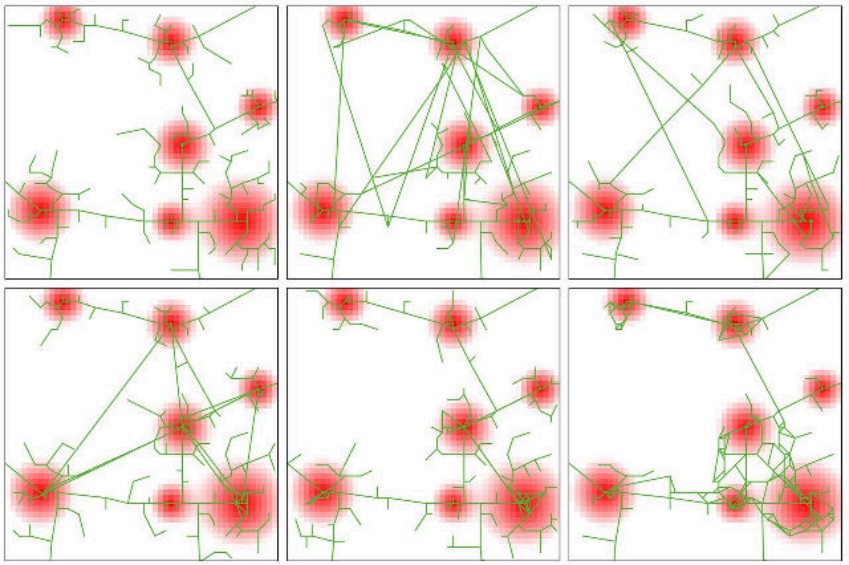

→ various applications to empirical data : railways and urban population, transport projects and real estate prices

# Causality regimes in co-evolution models

→ Simulation models with reciprocal interactions between networks and land-use at the micro scale effectively do produce co-evolution in the statistical sense at the macro-scale




*Urban system co-evolution model*
[Raimbault, 2021b]



*Urban morphogenesis co-evolution model*
[Raimbault, 2019]

# Extending the method with Information Theory

$\rightarrow$ causality in complex systems estimated with more robust and non-linear approaches, for example based on Information Theory for causality between scales by [Rosas et al., 2020]

$\rightarrow$ applied to urban systems multi-scale simulation models by [Raimbault and Pumain, 2023] and [Raimbault, 2026]

**Research question :**

*How can measures based on Information Theory, in particular Transfer Entropy, be used to extend the method to identify causality regimes and co-evolution?*

# Method

For two variables $X, Y$, Transfer Entropy computed for a delay $\tau$ as

$$TE_{X \to Y} = I(Y_{t+\tau}; X_t, Y_t) - I(Y_{t+\tau}; Y_t)$$

Effective Transfer Entropy is used to remove bias with bootstrapped surrogates :

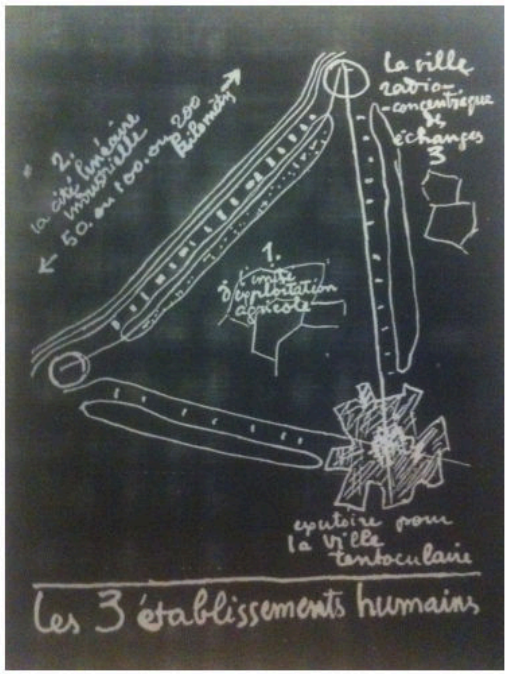
$$ETE_{X \to Y} = TE_{X \to Y} - \frac{1}{M} \sum_{m=1}^{M} TE_{X \to Y}^{(surr,m)}$$

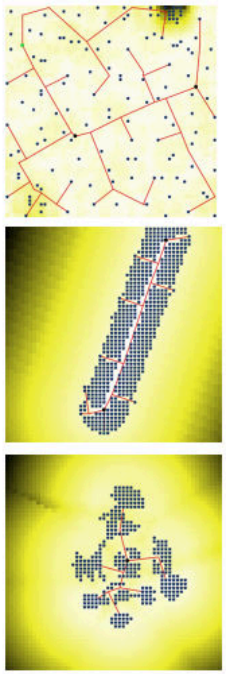
ETE is estimated on an ensemble with a Kraskov estimator (java jidt library implementation [Lizier, 2014]), on the population of urban objects

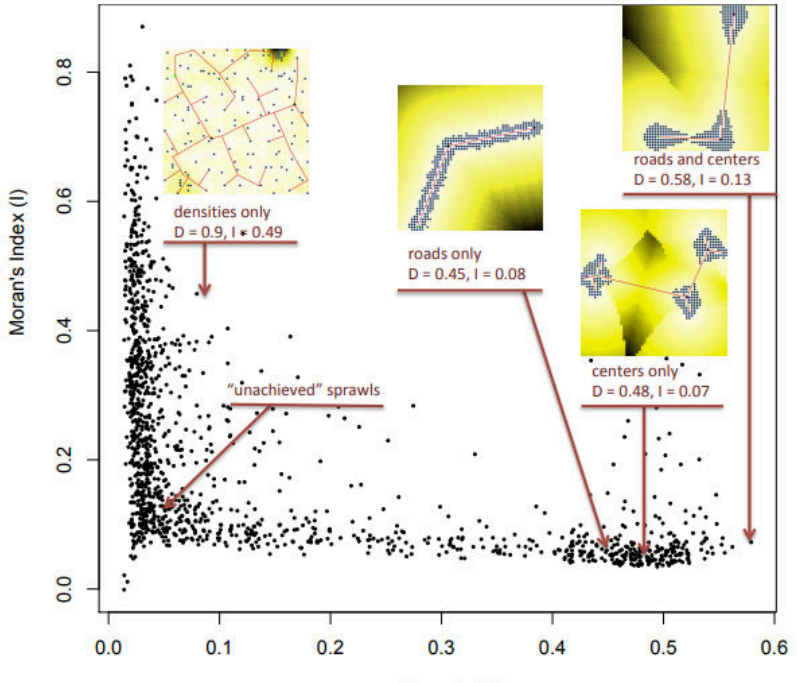
# Urban morphogenesis model

[Raimbault et al., 2014] : a simple and stylised urban morphogenesis model coupling urban development (cellular automaton) with a growing road network, in an iterative way:

- $N$ cells with the highest value (weighted average of density, distance to roads, distance to centre) are urbanised
- newly urbanised cells too far from the network are connected with a new road on the shortest path





# Implementation and experiments

→ models implemented in `NetLogo`, with the Transfer Entropy calculation from the `jidt` Java library integrated as an extension

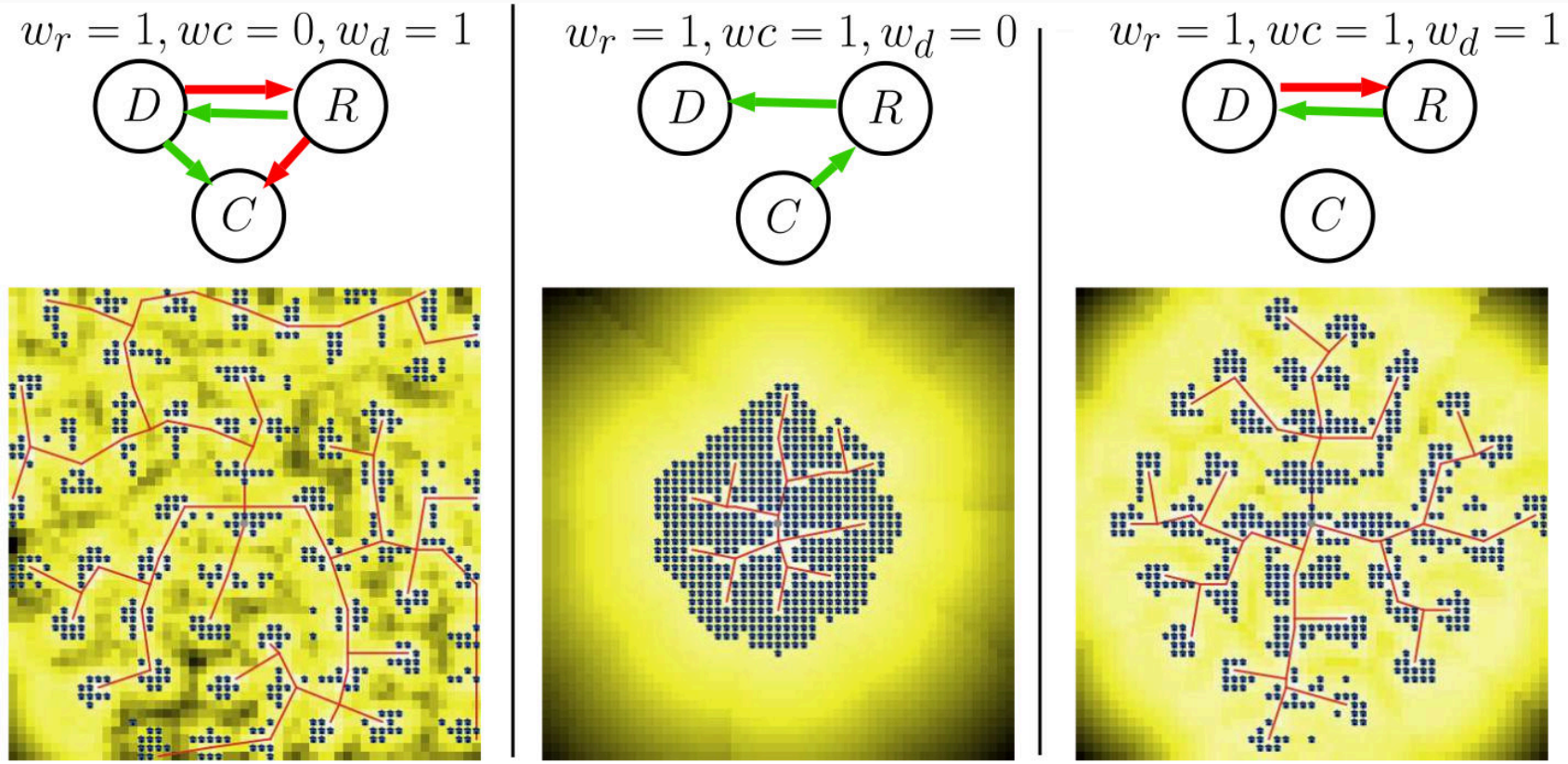
→ integration into the OpenMOLE model exploration and validation open source software [Reuillon et al., 2013] (*seamless (i) model embedding; (ii) access to HPC resources; (iii) exploration and optimisation algorithms*)




→ urban morphogenesis model run on synthetic systems of cities [Raimbault et al., 2019], with default model parameters

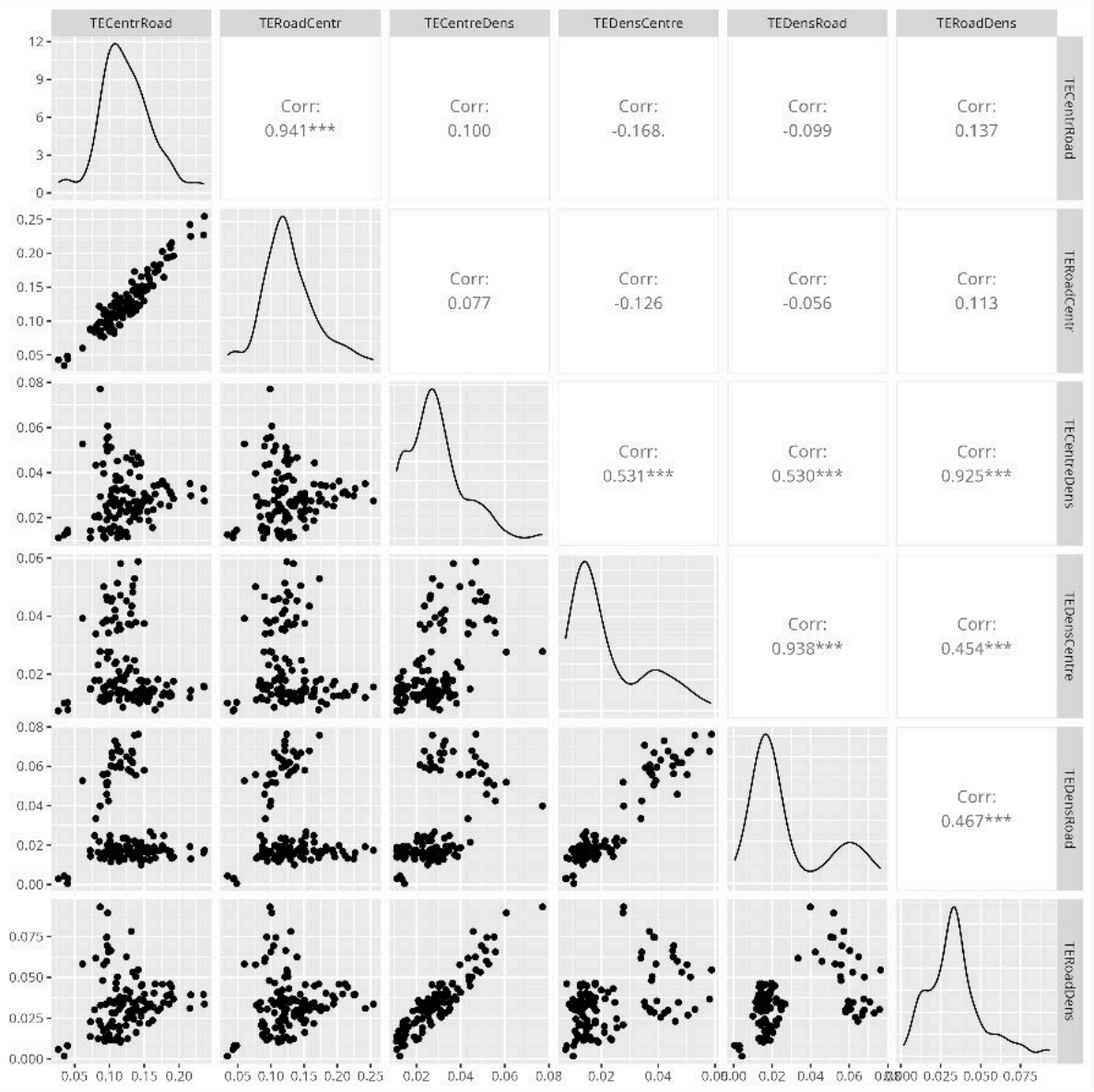
→ a grid parameter sampling (step 0.5) is done on weights parameters ($w_{density}$, $w_{roads}$, $w_{centre}$) with 10 repetitions ; ETE computed on all couples of cell value variables (density, distance to roads, distance to centre), with $1 \leq \tau \leq 3$

# Examples of causality regimes




*Examples of Granger causality regimes previously obtained in [Raimbault, 2021a]*

# Results with Transfer Entropy




Proportion of regimes, by number of reciprocal causalities (ETE in both directions larger than its median)

- 0 : 35%
- 1 : 33%
- 2 : 28%
- 3 : 4%

With lagged correlations

- 0 : 42%
- 1 : 34%
- 2 : 22%
- 3 : 1%

# Discussion

$\rightarrow$ a proof-of-concept of extending the causality regimes/coevolution method with Transfer Entropy

**Work in progress**

- Statistical validation, larger sampling, interpretation of regimes
- Pattern Search Exploration algorithm to explore the feasible space of causality regimes [Chérel et al., 2015]

**Future work**

- Test on empirical data
- Systematic benchmark of co-evolution models [Le Néchet and Raimbault, 2026]

Code and data available at

`https://github.com/JusteRaimbault/CoevolutionNwTerritories/tree/master/Models/MesoCoevolCA`

# References i


Blanquart, C. and Koning, M. (2017).
**The local economic impacts of high-speed railways: Theories and facts.**
*European Transport Research Review*, 9(2):12.

Chérel, G., Cottineau, C., and Reuillon, R. (2015).
**Beyond corroboration: Strengthening model validation by looking for unexpected patterns.**
*PloS one*, 10(9):e0138212.

Le Néchet, F. and Raimbault, J. (2026).
**Modelling coevolution between networks and space: a multiscalar approach.**
*La coévolution transport, mobilité, territoire*, pages 273–310.

# References ii


Lizier, J. T. (2014).
**Jidt: An information-theoretic toolkit for studying the dynamics of complex systems.**
*Frontiers in Robotics and AI*, 1:11.

Raimbault, J. (2017).
**Identification de causalités dans des données spatio-temporelles.**
In *Spatial Analysis and GEOmatics 2017*.

# References iii




Raimbault, J. (2019).
**An urban morphogenesis model capturing interactions between networks and territories.**
In *The mathematics of urban morphology*, pages 383–409. Springer.



Raimbault, J. (2021a).
**Characterising and modeling the co-evolution of transportation networks and territories.**
*arXiv preprint arXiv:2110.15950*.

# References iv


Raimbault, J. (2021b).
**Modeling the co-evolution of cities and networks.**
In *Handbook of cities and networks*, pages 166–193. Edward Elgar Publishing.

Raimbault, J. (2026).
**Quantifying emergence in systems of cities multiscale simulation models.**
In *CCS/France 2026*.

# References v




Raimbault, J., Banos, A., and Doursat, R. (2014).
**A Hybrid Network/Grid Model of Urban Morphogenesis and Optimization.**
In Aziz-Alaoui, M. A., Bertelle, C., Liu, X. Z., and Olivier, D., editors, *Proceedings of ICCSA 2014*, pages 51–60, Le Havre, France. Université de Normandie.



Raimbault, J., Cottineau, C., Le Texier, M., Le Néchet, F., and Reuillon, R. (2019).
**Space matters: extending sensitivity analysis to initial spatial conditions in geosimulation models. jasss.**

# References vi


Raimbault, J. and Le Néchet, F. (2021).
**Introducing endogenous transport provision in a luti model to explore polycentric governance systems.**
*Journal of Transport Geography*, 94:103115.

Raimbault, J. and Pumain, D. (2023).
**Innovation dynamics in multi-scalar systems of cities.**
In *Artificial Life Conference Proceedings 35*, volume 2023, page 143. MIT Press One Rogers Street, Cambridge, MA 02142-1209, USA journals-info ....



Reuillon, R., Leclaire, M., and Rey-Coyrehourcq, S. (2013). **Openmole, a workflow engine specifically tailored for the distributed exploration of simulation models.** *Future Generation Computer Systems*, 29(8):1981–1990.



Rosas, F. E., Mediano, P. A., Jensen, H. J., Seth, A. K., Barrett, A. B., Carhart-Harris, R. L., and Bor, D. (2020). **Reconciling emergences: An information-theoretic approach to identify causal emergence in multivariate data.** *PLoS computational biology*, 16(12):e1008289.